pilot | Making Research Data FAIR

indoc RESEARCH

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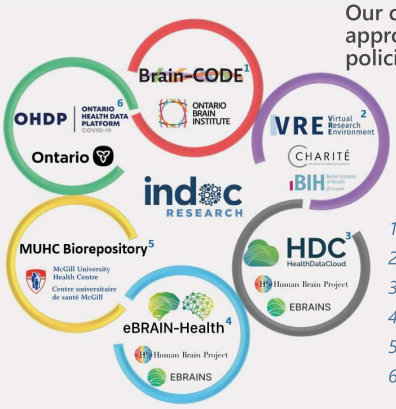
About Indoc

- Not-for-profit company established in Toronto, Canada in 2005
- Expansion of our operations in Europe with the creation of not-for-profit *Indoc Europe* in Mainz, Germany in 2021
- Vision: Create sustainable and interoperable data ecosystems to deepen our understanding of human health and disease
- Our Expertise
  - Software Development
  - Security and privacy
  - Research operations
  - Data & project management
  - Governance & data protection
  - Scientific domain expertise
- We build secure, scalable and extensible data platforms to support researchers in managing, integrating and analyzing complex medical data from diverse sources

Data Ecosystems by Indoc

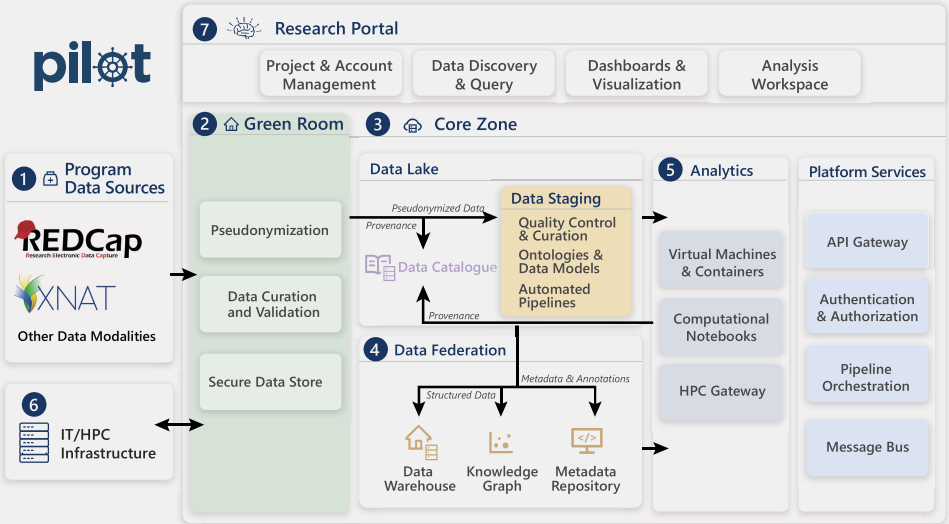
Our open-source & ecosystem approach ensures that knowledge, policies & technological advances developed for one program are available to the broader research community



1 - https://www.braincode.ca/
2 - https://tinyurl.com/2m7ucda9
3 - https://www.healthdatacloud.eu/
4 - https://tinyurl.com/4d8hw3hx
5 - https://cvsignature.ca/
6- https://ohdp.ca/

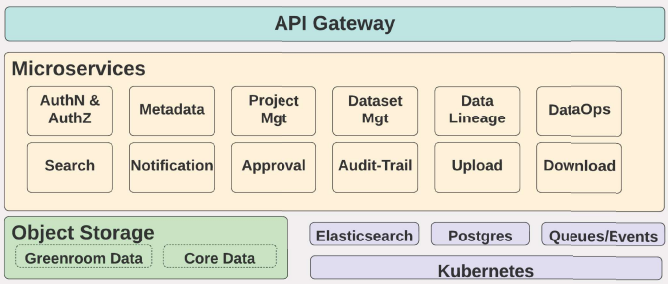
Pilot

- Pilot is an open-source data platform developed by Indoc making research data Findable, Accessible, Interoperable & Reusable (FAIR)
- Pilot can be deployed on compute clouds, on-premise infrastructure, or as a hybrid
- Pilot supports data federation across multiple centres enabling data discovery while complying with data residency requirements



1. Support diverse data types of different modalities and sensitivities such as electronic health records (EHRs), radiological images, genomic sequences
2. Green Room is a landing zone for any data coming into the Platform. Highly sensitive data such as EHRs can be pseudonymized and transformed before being made available for research
3. The Core Zone receives data processed in the Green Room for analysis by all project collaborators
4. Data standardization and annotation based on ontologies to support FAIR
5. Workspace with analysis & visualization tools such as Jupyter notebook
6. Integration with high-performance computing infrastructure for executing resource intense workflows
7. Web Portal for administering the platform, managing users & data, and accessing analysis workspaces

Pilot Architecture



- Following a microservices architecture, the Pilot platform is composed of small services each handling discrete tasks
- Each service, containerized with all its software dependencies, can be developed, deployed and scaled independently
- By utilizing Kubernetes to manage the containers, Pilot is portable to a variety of environments including commercial clouds and OpenStack

Governance & Data Protection

- Legislative scope
- Privacy & data protection impact assessments

- Governance policies
- Approval & oversight of platform users

- Technical & organization measures
- Threat, risk & vulnerability assessments

Indoc participated in a successful GDPR legal evaluation of the Charité Virtual Research Environment

Contact Us

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